

RIO CUARTO, Argentina

WMO: 874530

Lat: 33.0956S

Lon: 64.2769W

Elev: 421

StdP: 96.37

Time Zone: -3.00 (W03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	0.2	1.8	-10.2	1.7	10.4	-8.0	2.0	9.8	14.6	15.0	12.9	14.8	2.6	270	0.525

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	10.7	34.0	20.7	32.4	20.4	31.1	20.1	23.9	29.6	22.9	28.5	22.1	27.7	7.3	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
22.2	17.8	26.6	21.2	16.7	25.8	20.3	15.8	24.9	74.2	29.7	70.4	28.6	67.2	27.8	28.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.3	11.9	10.4	DB	-3.0	37.5	2.3	2.0	-4.6	38.9	-6.0	40.1	-7.3	41.2	-9.0	42.7	(4)
(5)			WB	-4.3	26.0	2.0	1.1	-5.8	26.8	-6.9	27.5	-8.1	28.1	-9.5	29.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	17.0	23.4	22.1	20.4	17.0	13.6	10.5	9.9	12.0	14.6	17.7	20.6	22.7	(6)	
	DBStd	5.85	2.89	3.28	3.32	3.76	3.65	3.05	3.53	4.19	4.21	3.93	3.57	3.25	(7)	
	HDD10.0	119	0	0	0	2	8	29	45	25	9	1	0	0	(8)	
	HDD18.3	1151	2	7	18	68	153	234	261	203	124	59	18	4	(9)	
	CDD10.0	2676	414	338	322	212	121	46	43	86	147	239	317	393	(10)	
	CDD18.3	668	158	111	81	28	7	0	1	6	13	39	85	139	(11)	
	CDH23.3	6046	1405	892	611	229	42	3	9	88	163	423	843	1338	(12)	
(13)	CDH26.7	2191	551	316	182	47	4	0	2	27	44	144	317	557	(13)	
(14)	Wind	WSAvg	4.8	4.8	4.5	4.4	4.3	3.9	4.5	5.1	5.6	5.8	5.5	5.3	(14)	
(15) Precipitation	PrecAvg	808	130	98	108	55	29	16	12	14	40	83	122	137	(15)	
	PrecMax	1131	295	356	465	142	218	111	113	85	156	266	232	325	(16)	
	PrecMin	467	11	23	10	2	0	0	0	0	0	2	36	27	(17)	
	PrecStd	152	65	66	74	35	37	20	17	18	34	50	55	63	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.3	34.6	32.8	30.7	27.0	23.2	24.1	30.0	30.9	33.8	34.5	36.1	(19)	
		MCWB	21.4	21.9	21.1	19.9	18.2	14.2	14.4	17.1	16.7	19.6	19.2	21.6	(20)	
	2%	DB	33.2	32.1	30.7	27.9	24.2	20.3	20.9	25.5	27.4	30.2	32.1	33.8	(21)	
		MCWB	21.6	22.0	21.2	18.8	16.9	12.6	12.7	14.7	15.5	17.5	18.6	20.7	(22)	
	5%	DB	31.5	30.2	28.8	26.1	22.0	18.3	18.7	22.1	25.0	27.8	30.2	31.9	(23)	
		MCWB	21.3	21.5	20.4	17.6	15.4	11.5	11.2	12.3	14.4	16.6	18.1	20.0	(24)	
	10%	DB	29.8	28.4	27.0	24.1	19.9	16.6	16.3	19.8	22.6	25.2	28.2	29.9	(25)	
		MCWB	20.8	20.9	19.7	17.1	14.5	10.8	10.1	11.5	13.1	15.2	17.3	19.5	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	25.1	25.1	24.2	22.6	19.7	17.0	16.9	18.7	19.2	21.5	22.4	24.1	(27)	
		MCDB	30.0	30.8	29.8	27.1	24.2	20.8	21.6	26.4	26.9	27.7	28.9	31.6	(28)	
	2%	WB	23.7	23.7	22.7	20.8	18.2	14.8	14.5	16.2	17.1	19.5	20.8	22.7	(29)	
		MCDB	29.3	28.7	27.5	24.4	22.5	17.5	17.8	22.7	23.4	26.5	27.5	29.9	(30)	
	5%	WB	22.7	22.7	21.6	19.4	16.9	13.4	12.9	14.2	15.7	18.2	19.7	21.6	(31)	
		MCDB	28.7	27.3	26.4	23.8	20.3	16.0	15.8	19.6	22.2	24.5	26.7	28.5	(32)	
	10%	WB	21.8	21.8	20.5	18.2	15.6	12.0	11.4	12.4	14.4	16.9	18.6	20.7	(33)	
		MCDB	27.8	26.6	25.2	22.5	18.5	15.0	15.0	18.4	20.4	23.0	25.4	27.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	10.7	10.1	10.3	10.4	9.9	10.7	11.0	12.0	12.1	11.6	11.8	11.5	(35)	
		MCDBR	13.2	12.5	12.7	13.8	12.9	14.1	14.9	16.3	15.9	15.3	15.0	14.4	(36)	
	5% WB	MCWBR	5.5	5.1	5.6	6.4	6.7	7.8	8.1	7.9	7.5	6.7	6.1	5.7	(37)	
		MCWBR	11.3	10.4	10.9	10.9	10.3	10.8	11.3	13.9	13.0	12.7	12.6	12.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.378	0.362	0.362	0.330	0.310	0.322	0.356	0.404	0.389	0.380	0.385	(40)	
		taud	2.374	2.407	2.453	2.418	2.459	2.515	2.441	2.325	2.189	2.293	2.352	2.366	(41)	
	Ebn at Noon		952	939	918	859	837	834	836	849	858	917	952	957	(42)	
		Edh at Noon	129	121	108	100	85	75	85	107	138	134	132	132	(43)	
(44) All-Sky Solar Radiation	RadAvg		7.25	6.24	5.24	3.83	2.83	2.63	2.83	3.77	4.90	5.82	6.90	7.38	(44)	
	RadStd		0.44	0.49	0.46	0.47	0.31	0.29	0.23	0.32	0.42	0.54	0.44	0.49	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	+0.62	+0.50	+0.73	N/A	N/A	N/A	N/A	(46)
(47)	Regional (5 neighbors)	N/A	N/A	N/A	+0.56	+0.62	+0.78	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page